

Toward A Framework for Ethical Analysis

Social Issues & Professional Practice in
IT & Computing
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Introduction to Ethical Issues

Motivating Scenarios

Scenario 1: When is it justifiable to break the law?

- Example: Bad laws, easy-to-break laws.

Scenario 2: Should a company mine data without customer consent?

- Example: Tisetso uses data mining tools for marketing purposes.

Ethical considerations:

Is it ethical to use customer data for purposes not agreed upon?

Key questions to ask:

1. Should customers be notified?
2. Is there a need for a policy regarding data usage?

Scenario 1: Should I copy software?

Rajesh invests small amounts on the stock market. Last year he bought and successfully employed a software package to help him with his investments. Recently, he met Fundiswa who was also interested in using the software. Fundiswa borrowed the package, copied it and then returned it. Both vaguely knew that the software was proprietary but did not read up the details. Did Rajesh and Fundiswa do anything wrong, if so what?

More generally, try to answer the following related questions:

- Should the software package have been lent?*
- When is it justifiable to break the law? Bad law, inappropriate law or if the law is easy*
- to break?*

Scenario 2: Should a company mine data?

Consider the case where Tisetso sells hardware and software to over 100 000 customers per year. She has 10 years of experience. As part of the billing process she keeps information on customers. She buys and uses a data mining tool to derive useful information about her client's information such as postal codes, credit card numbers, ID numbers, etc.

Most of this information identifies groups and not individuals. She can then use the information to market her wares more efficiently.

- Is this ethical since customers did not give her the information for this purpose?*
- Should the customer be notified?*
- Is there a need for establishment of a policy? If so, what should this policy look like?*

Professional responsibility (professional Ethics):

Do professionals have a responsibility to ensure computing serves humanity well?

Data Mining

***Data mining** is a process of exploration and analysis of large quantities of data, by automatic or semi-automatic means. This is done in order to discover meaningful patterns and rules. In many cases, the data was not collected primarily for the purpose that the data is used for.*

Scenario 3: Freedom of Expression

A student, Gert, posts sex fantasies on a blog (alike blogger.com), called Ling's Journey. The stories are fictional, but Gert named the main character, Ling, after a real student. In the story, he described the rape, torture and murder of Ling. He is also member of a social media newsgroup (alike Facebook), discussing sex acts. An alumnus saw this and reported it to the University as well as to the social media companies. Gert was then arrested and held in custody and the social media companies suspended his account. He was charged with transmitting communication of threat to injure another person, and violating community policies of the respective social media platforms.

- The charges were eventually dropped and his accounts reinstated. Did Gert really do anything wrong?*

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- Should self-censorship be enforced. Who decides what is acceptable?*
- Is the social media software company responsible for the content on their site?*
- Should this data—the blog posts and the social media newsgroup chats—be kept online accessible indefinitely?*
- Is there a need for a public policy on freedom of expression on social media, as compared to, say, printed hardcopy books and magazines?*

Scenario 4: Professional Responsibility

Khadeejah works for a software development company that develops computer games for children aged 8-14. The latest game that Khadeejah worked on, uses inferential reasoning on social roles and allows players to choose different characters, primarily a macho man or a sexy woman. The game is used mainly by boys. Recently, Khadeejah attended a conference on gender and marginalized groups, where she described the above.

The conference delegates discussed the issue of lower participation of women in computing and how to make the industry more attractive to women. At the conference, she also learned that recent research has shown that more diverse teams produce better results, mainly because assumptions are not taken for granted anymore and more ideas are brought to the table.

Back at work, Khadeejah realized that her production team is mostly male.

- Should she refuse to work on this team? Should she ask for the team to be reviewed?*
- Do you think that the game will sell as well if a different message was given?*
- What is Khadeejah's responsibility, if any, in the team and regarding the games' topics?*
- Should the message in games be taken into account?*
- Should software development teams be diverse? It justifiable to refuse to work in a team that is not diverse?*
- Is diversity, or a lack thereof, a responsibility of the profession?*

Reasoning

Automated reasoning is a way to infer implicit knowledge from explicitly represented information, using the rules of inference together with a logic in which the information is represented and a set of algorithms that automate and scale up this process on the computer.

Ethical Framework in IT

****Professional Responsibility in IT**

IT professionals must ensure computing serves humanity well.

Ethical issues in IT directly impact **society**, **business**, and **personal freedom**.

****The Role of IT Ethics****

Ethical decision-making in IT ensures fairness, privacy, and transparency.

Ethical dilemmas arise when professional responsibilities **conflict with business or personal interests**.

Components of Ethical Analysis Framework

****Identification of Ethical Issues****

Recognize problems such as privacy breaches, misuse of data, and bias in algorithms.

****Stakeholder Analysis****

Who is affected by the ethical decision? (e.g., customers, employees, society).

****Consequences of Actions****

Consider long-term and short-term outcomes of a decision on various stakeholders.

****Application of Ethical Principles****

Utilize ethical theories such as Utilitarianism (greatest good for the greatest number) or Deontology (duty-based ethics).

Ethical Decision-Making in Practice

****Real-World Example****

Data mining without explicit customer consent: **How can ethical analysis be applied?**

Example: Should a company use customer data for targeted marketing if it was not explicitly collected for that purpose?

****Critical Thinking and Ethical Responsibility****

IT professionals must evaluate both legal and moral implications in their decisions.

Importance of transparency, honesty, and fairness in decision-making.

Key Ethical Challenges in IT

****Privacy Concerns****

How to protect individual privacy in an age of data collection and surveillance?

Example: Social media platforms collecting data for targeted ads without clear consent.

****Intellectual Property and Ownership****

Ethical challenges in software development, digital content creation, and copyright violations.

Example: Sharing pirated software or using copyrighted material without permission.

****Transparency and Accountability****

Ensuring decision-making processes in IT systems are clear, fair, and open.

Example: AI systems making biased decisions without transparency about the algorithms used.

Framework Application

****Step-by-Step Ethical Framework****

1. Identification of the Ethical Issue

Clearly define the dilemma (e.g., privacy violation).

2. Stakeholder Analysis

Understand who is affected by the decision (e.g., customers, society).

3. Evaluating Consequences

Consider all potential outcomes of your actions (e.g., damage to reputation, legal ramifications).

4. Application of Ethical Principles

Use ethics theories to justify your decisions (e.g., utilitarianism).

Example Case Study:

Application of the framework to a data mining dilemma in business.

Challenges in Ethical Decision-Making

****Conflicting Stakeholder Interests****

Balancing business goals (e.g., profit maximization) with ethical considerations (e.g., data privacy).

Example: Companies selling personal data to third parties for marketing purposes.

****Uncertainty in Complex Cases****

Some ethical issues don't have clear answers, especially with emerging technologies like AI and automation.

Example: How to ensure AI systems are ethical when there are no clear regulations in place?

Conclusion

****Summary****

Ethical analysis is essential for responsible decision-making in IT.

A **structured ethical framework** can help address dilemmas in technology development.

****Final Thought****

- **Ethical analysis** is a continuous, evolving practice as technology advances.
- **IT professionals** must stay **vigilant** and **proactive** in applying ethical reasoning to their work.